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1 Foundations and Large Cardinals

This part is self-contained.
The goal is to prove axiom of limitation of size using small amount of simple axioms and definitions.

Axiom 1. Axiom of limitation of size:

$$(K \in V) \longleftrightarrow (|K| < |V|)$$

Here $|\cdot|$ is a functional symbol defined as following.

Definition 1.1. (cardinality of a class)

$$|K| := \mathrm{Ord} \cap \bigcap \{a \in \mathrm{Ord} | K \preccurlyeq a\}$$

Definition 1.2.

$$(X \preccurlyeq Y) \longleftrightarrow (\exists f : (f : X \to Y) \wedge (f \text{ injective}))$$

The intended behaviour of this definition is to have cardinality of argument if the argument of a functional symbol is a set, and Ord if the argument is a proper class.
We will introduce some axioms to make such behaviour provable.

Axiom 2.

$$|\cdot| : V \to \mathrm{Ord}$$

Actually, since $|\cdot|$ is a functional symbol, we mean the following:

$$(x \mapsto |x|) : V \to \mathrm{Ord}$$

Functional symbol is defined on any class as argument, while $(x \mapsto |x|)$ is defined on sets only. By definition, $(x \mapsto |x|)$ is a class function defined as a class of pairs $\{\langle x, |x| \rangle | x \text{ any set}\}$.

Actually, we can use the following axiom:

Axiom.

$$(x \text{ set}) \longrightarrow (|x| \in \mathrm{Ord})$$

or even

Axiom.

$$(x \text{ set}) \longrightarrow (|x| \text{ set})$$

since later we will prove that

Theorem.

$$(x \text{ set}) \longrightarrow (|x| \in \text{Ord})$$

Axiom 3. Sets are hereditary:

$$(A \subseteq B) \wedge (B \text{ set}) \longrightarrow (A \text{ set})$$

Some equivalent(in normal context) statements:

$$(K \in V) \longleftrightarrow (|K| \subsetneq |V|)$$

$$(K \in V) \longleftrightarrow (|K| \neq |V|)$$

because $|V| = \text{Ord}$ and

Theorem 1.1.

$$|V| = \text{Ord}$$

Proof.
Done.

In this text we will consider the simplest definition of ordinals which assumes that axiom of regularity holds. It is done for shortness, and can be done with a definition that doesn't require the regularity axiom.

Definition 1.3. predicates on classes "is transitive" and "is ordinal"

$$tr(X) \longleftrightarrow (\forall Z \in X : Z \subseteq X)$$

$$ord(X) \longleftrightarrow tr(X) \wedge (\forall Y \in X : tr(Y))$$

the other possibilities are

$$ord(X) \longleftrightarrow tr(X) \wedge (\text{E Wo } X)$$

$$ord(X) \longleftrightarrow tr(X) \wedge (\text{E Wo } X) \wedge (\text{E Fr } X)$$

etc ...

$$(X \text{ set}) \longrightarrow$$

1.1 Proof of the main statement

We want to prove the following statement:

$$(K \in V) \longleftrightarrow (|K| \neq |V|)$$

Firstly, we will prove some lemmas:

Lemma 1.1.

$$\text{Ord} \notin \text{Ord}$$

Proof.
Assume Ord $\in$ Ord.
Therefore $\in$ well-orders Ord as a well-founded strict linear order.
In particular strict order is irreflexive.
Since using assumpton, Ord is an element of Ord, by irreflexetivity, we have Ord $\notin$ Ord. Contradiction with assumption. Therefore Ord $\notin$ Ord is proven.
Done.

Lemma 1.2.

$$|V| = \text{Ord}$$

Proof.
...
Done.

Lemma 1.3. (1st part of the statement, from left to right)

$$(K \in V) \longrightarrow (|K| \neq |V|)$$

Proof.
Assume $(K \in V)$.
By our axiom, since $(K \in V)$ we have $(|K| \in \text{Ord})$.
By previous lemmas we have $|V| = Ord$ and Ord $\notin$ Ord.
From these three statements we deduce that $|K| \neq |V|$.
Done.

Lemma 1.4. (2nd part of the statement, from right to left)

$$(|K| \neq |V|) \longrightarrow (K \in V)$$

Proof.
Assume $(|K| \neq |V|)$.
We know that $(|V| = \text{Ord})$, so $(|K| \neq Ord)$.
Recalling the definition: $|K| := \text{Ord} \cap \bigcap\{a \in \text{Ord} | K \preccurlyeq a\}$.
Now we have that there exists some $m \in$ Ord, which is not in $\bigcap\{a \in \text{Ord} | K \preccurlyeq a\}$. (vice versa situation is not possible since any non-emptyintersection of subclasses of Ord is subclass of Ord)
...
Done.